Answers

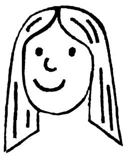
Worksheet

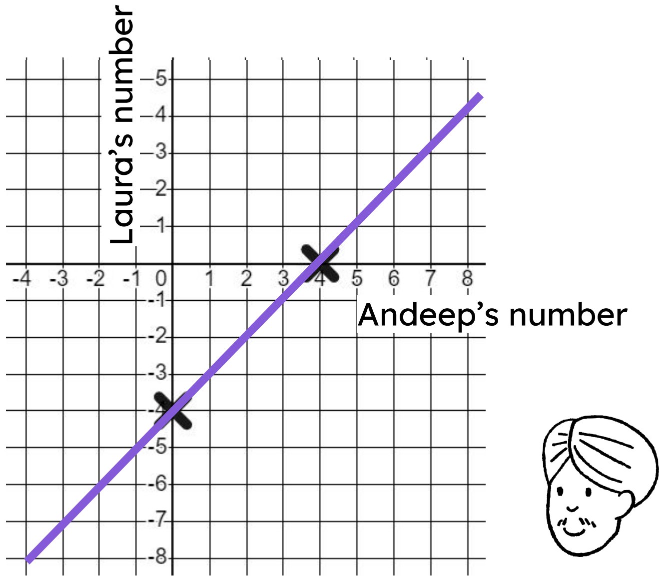
Checking and securing plotting a relationship

Task A: Checking understanding of plotting a coordinate

1) Laura says...

Whatever number Andeep chooses I'm going to subtract 4 from it.



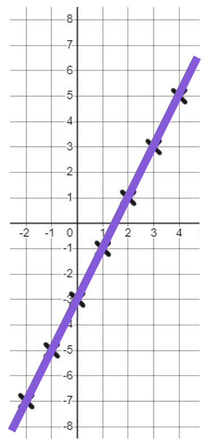


Plot some points to graphically represent this relationship.

2) Complete the tables of values and plot these relationships.

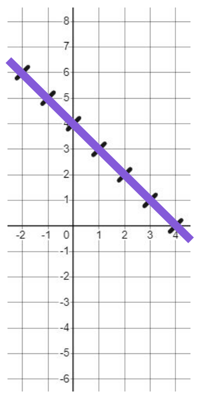
a) $y = 2x - 3$

x	-2	-1	0	1	2	3	4
y	-7	-5	-3	-1	1	3	5



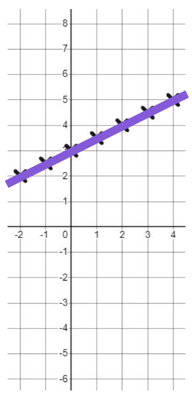
b) $x + y = 4$

x	-2	-1	0	1	2	3	4
y	6	5	4	3	2	1	0



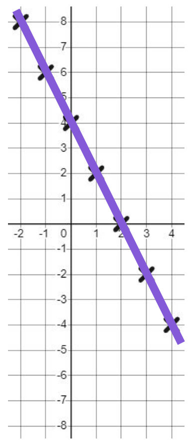
c) $y = \frac{1}{2}x + 3$

x	-2	-1	0	1	2	3	4
y	2	2.5	3	3.5	4	4.5	5



d) $y = 4 - 2x$

x	-2	-1	0	1	2	3	4
y	8	6	4	2	0	-2	-4

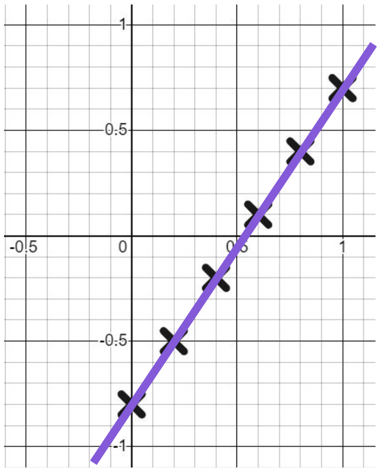


3) Complete the tables of values for this relationship.

$y = 1.5x - 0.8$

x	0	0.2	0.4	0.6	0.8	1
y	-0.8	-0.5	-0.2	0.1	0.4	0.7

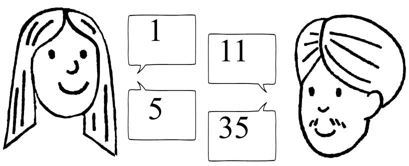
Draw suitable axes to graphically represent the coordinates.



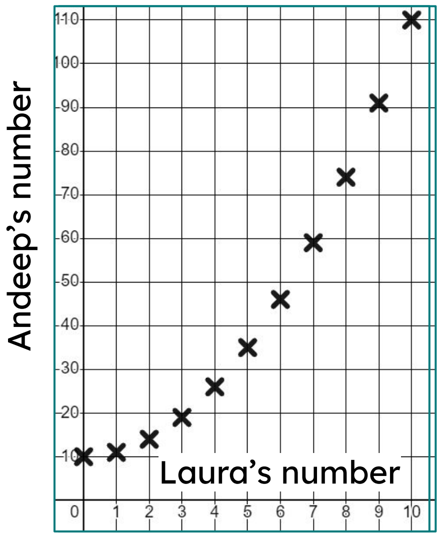
Task B: Securing understanding of plotting a relationship

1) This time Andeep is squaring Laura's number and adding 10.

Laura (L)	1	2	3	4	5	6	7	8	9	10
Andeep (A)	11	14	19	26	35	46	59	74	91	110



a) Complete the table of values and plot the relationship.



b) Explain why $(5,25)$ is not on this curve.

It does not fit the relationship. $5^2 + 10 \neq 25$

c) Write this relationship algebraically. $A = L^2 + 10$

2) This time our Oak children are sharing 24 apples.

a) Complete the table of values and plot the relationship.

People (p)	1	2	3	4	6	8	12	24
Apples (a)	24	12	8	6	4	3	2	1

b) Write this relationship algebraically.

$$\frac{24}{p} = a$$